expirement
friction force

expirement (frictional force)

part(1): find Static frictional coefficient (Ms)

أسماء: سحاق محمد الفواضلة

mass (Kg)	normal force (N) n: mg (g:10)	Fs (Newten)
0.200	2	1.2
0.220	2.2	1.32
0.240	2.4	1.44
0.260	2.6	1.56
0.280	2.8	1.68

$$\text{Slope} : \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope} : \frac{\Delta Fs}{\Delta \text{normal}} = \frac{Fs_2 - Fs_1}{n_2 - n_1}$$

$$\text{Slope} : \frac{1.68 - 1.2}{2.8 - 2}$$

Slope: 0.6 (without unit)

Slope: (Ms) : 0.6

allowed values (more than zero to less or equals to one)

part (2):

find the Kinetic frictional coefficient (MK)

mass (Kg)	normal force (N) n: mg (g:10)	FK (Newten)
0.200	2	0.8
0.22	2.2	0.88
0.24	2.4	0.96
0.26	2.6	1.04
0.28	2.8	1.12

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope} : \frac{\Delta FK}{\Delta \text{normal}} = \frac{FK_2 - FK_1}{n_2 - n_1}$$

$$\text{Slope} : \frac{1.12 - 0.8}{2.8 - 2}$$

Slope: 0.4 (without unit)

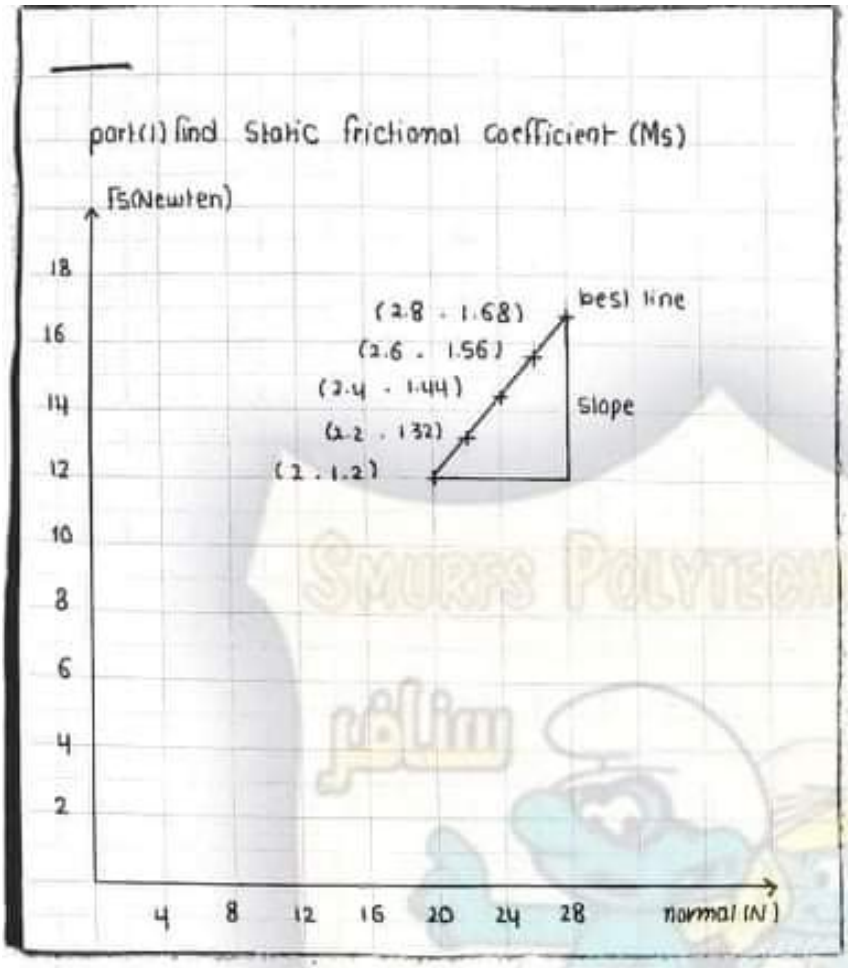
Slope: (MK) : 0.4

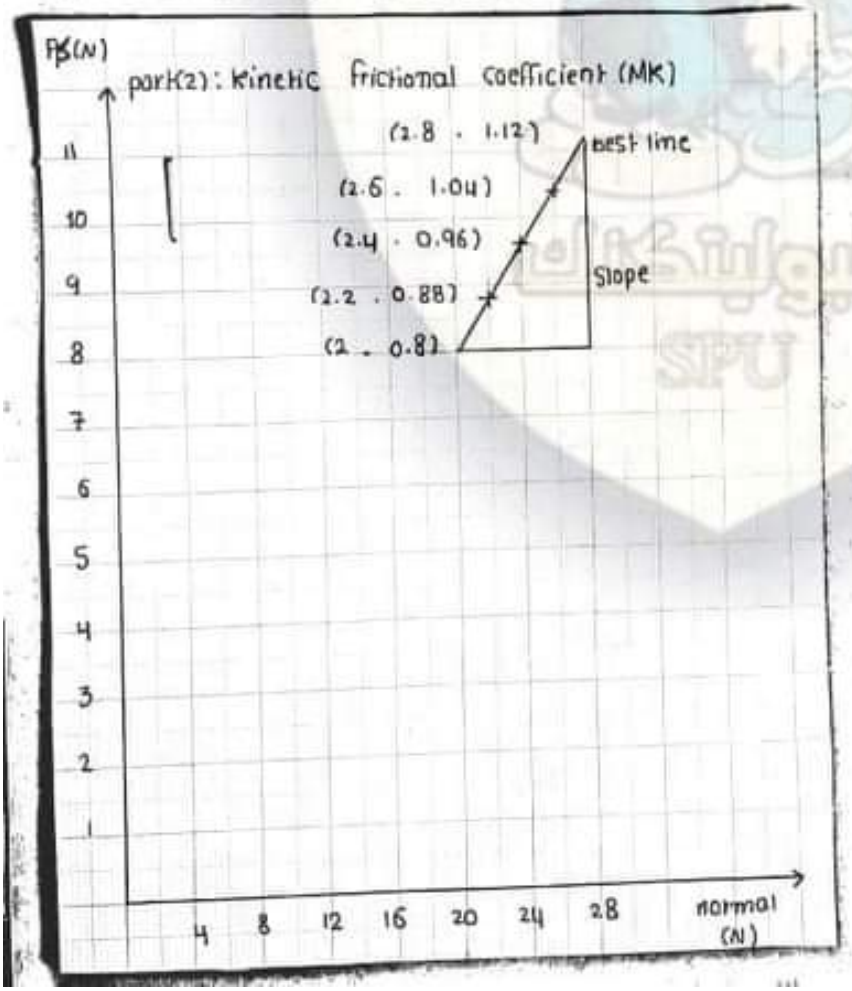
allowed values (more than zero to less or equals to one)

note that: (Kinetic frictional coefficient must be less than the static one!)

أسماء إسحاق محمد الفواضلة

part(1) find Static frictional coefficient (Ms)





expirement

simple pendulum

expirement (the Simple peandulum)

Case (1) to find (g)

أسماء إسحاق محمد الغواضنة

Length (meter)	time for tenth (t10) (sec)	T (time for one vibration (t10 /10)	$(T)^2$
0.2	8.97	0.897	0.8046
0.3	10.99	1.099	1.2078
0.4	12 .69	1.269	1.610
0.5	14. 19	1.419	2.013
0.6	15.54	1.554	2.4149
0.7	16.78	1.678	2.815
0.8	17.94	1.794	3.218

$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\text{Slope} = \frac{0.6 - 0.4}{2.41 - 1.61} = 0.25$$

$$\text{Slope} = \frac{g}{4\pi^2}$$

$$g = (\text{Slope})(4\pi^2)$$

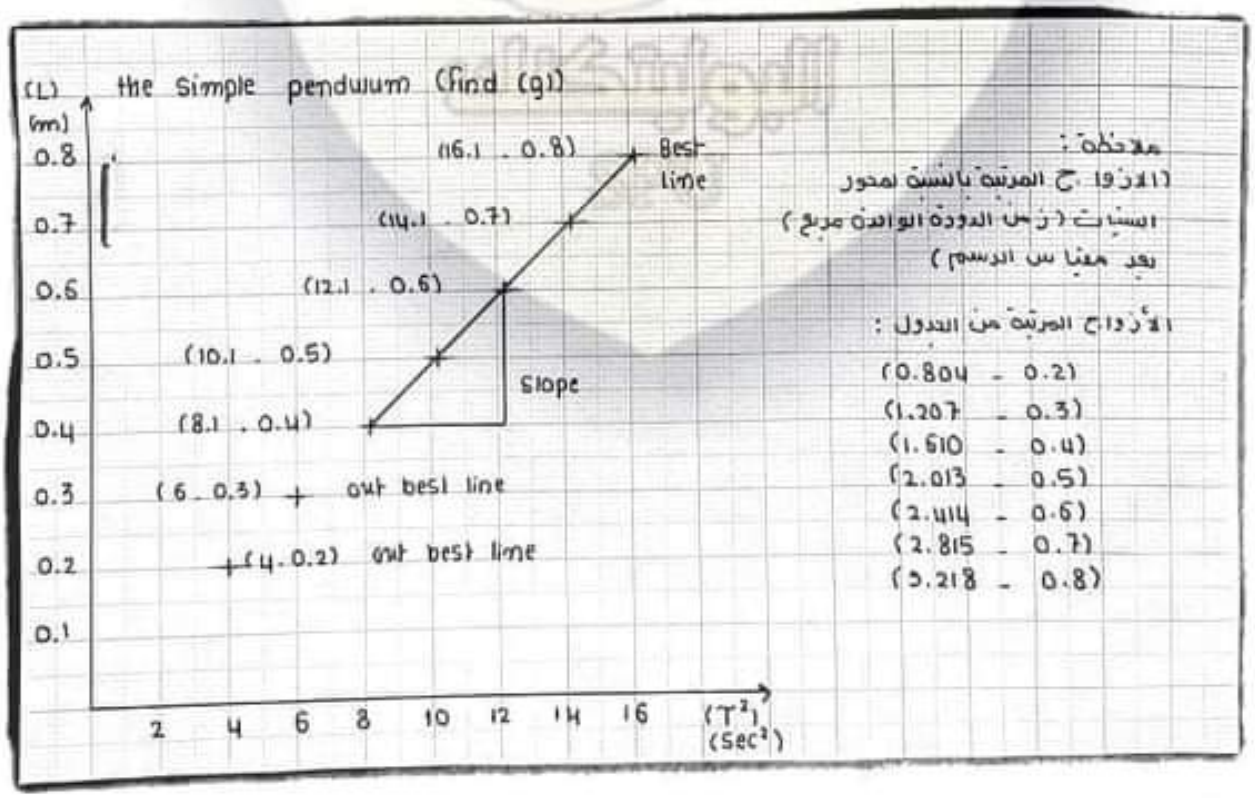
$$g = (0.25)(4)(3.14)^2$$

$g = 9.85$ meter/sec^2 (experemntal value)

Bec. we have true and exp values find % error

$$\text{Persentage Error} = \frac{|\text{true} - \text{exp.}|}{\text{true}} \times 100\%$$

$$\text{Persentage Error} = \frac{|9.8 - 9.851|}{9.8} \times 100\% = 0.5$$



expirement

hook’s law

Experiment (Hooke's Law)

Case (1): find Spring constant

Suppose that $x_0 = 0.07$ meter

أسماء إسحاق محمد الفواضلة

mass (kg)	x_i (m)	$F = mg$ (N)	$\Delta x = x_i - x_0$	$K = \frac{F}{\Delta x}$
0.06	0.13	0.6	0.06	10
0.07	0.14	0.7	0.07	10
0.08	0.15	0.8	0.08	10
0.09	0.16	0.9	0.09	10
0.1	0.17	1.0	0.1	10
0.11	0.18	1.1	0.11	10

$$\text{Slope} = \frac{\Delta F}{\Delta(\Delta x)} = \frac{1 - 0.7}{0.1 - 0.07}$$

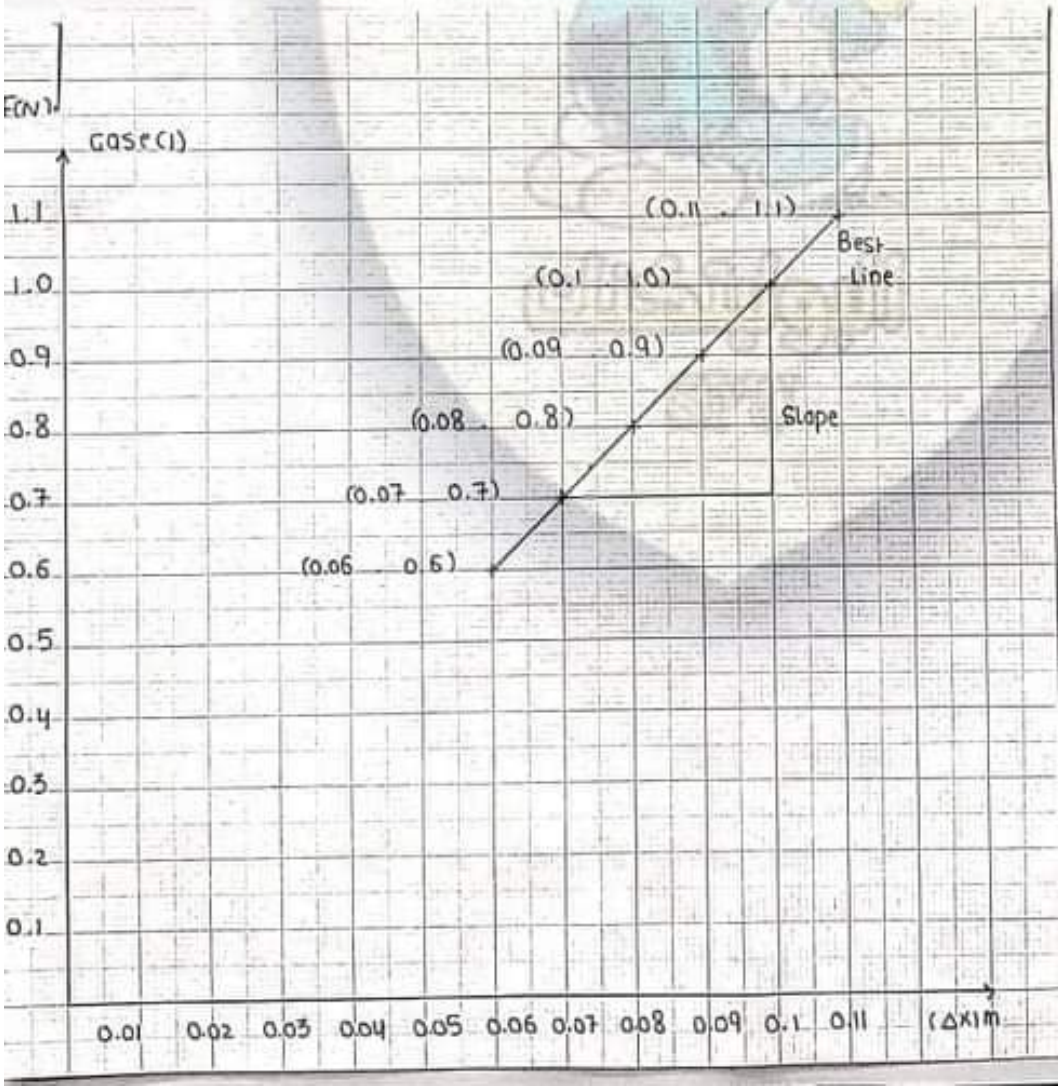
Slope = (K) = 10 N/meter

Standard deviation: zero

Because there is no difference between the values and average

$$K = \frac{\Sigma K}{n}$$

$K = 10 N/m$



expirement (HOOK's Law)

Case (2) : find Spring Constant

أسماء اسحاق محمد الفواضنة

mass (Kg)	time for tenth T_{10}	time for one	T^2/sec^2	جربت لمنزلين
0.06	4.86	0.486	0.2361	0.24
0.07	5.25	0.525	0.2756	0.28
0.08	5.62	0.562	0.3158	0.32
0.09	5.96	0.596	0.3552	0.36
0.1	6.28	0.628	0.3943	0.39

the slope is Spring Constant

$$\text{Slope} = \frac{\Delta \text{mass}}{\Delta T^2}$$

$$\text{Slope} = \frac{0.36 - 0.24}{0.09 - 0.06}$$

Slope = 0.25 Kg/sec²

